CS50

Programming basiacs D1

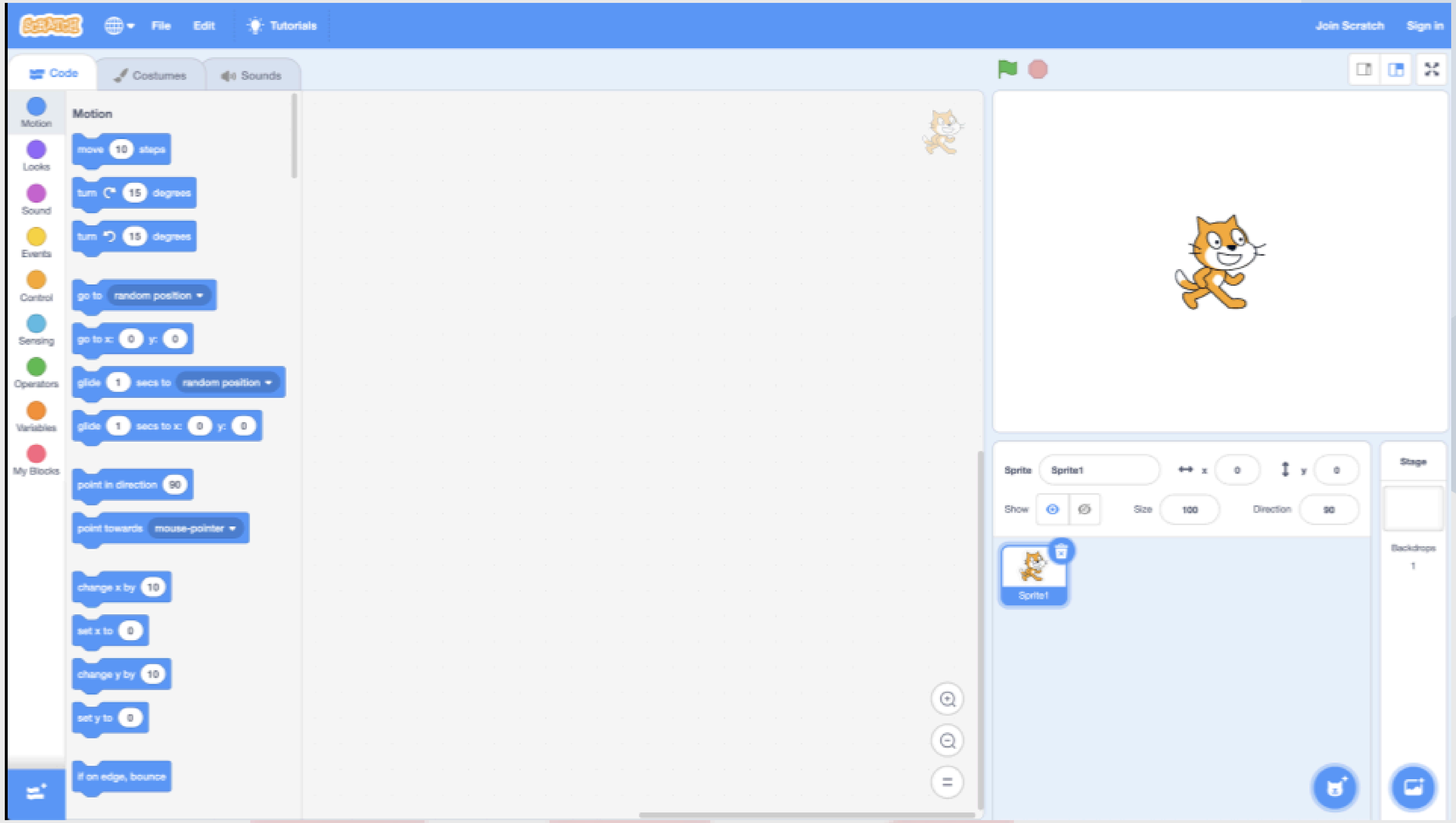
C++

Outlines:

1- Introduction.
2- First C++ program.
3- Variables.
4- Conditionals.

From Scratch to C++ programming:

Scratch to C: *After learning Scratch, a visual programming language, we're now moving to C programming. The concepts you learned in Scratch will still be useful here.*



اتعلمنا في scratch انها visual programming
language يعني بنستخدم blocks كل بلوك ليه
استخدام وكنا بنجمع مجموعة من الblocks مع
بعض علشان ننفذ أفكارنا وكمان اتعلمنا بعض
مفاهيم البرمجة زي conditions وloop
وvariable وdefine block وطبقنا كل المفاهيم
دي بإستخدام scratch
هنبدأ نتعلم لغة برمجة ++C علشان برضو نعرف
نتكلم مع الكمبيوتر وينفذ افكارنا وهنطبق كل
المفاهيم الي اتعلمناها في scratch بلغة ال++C

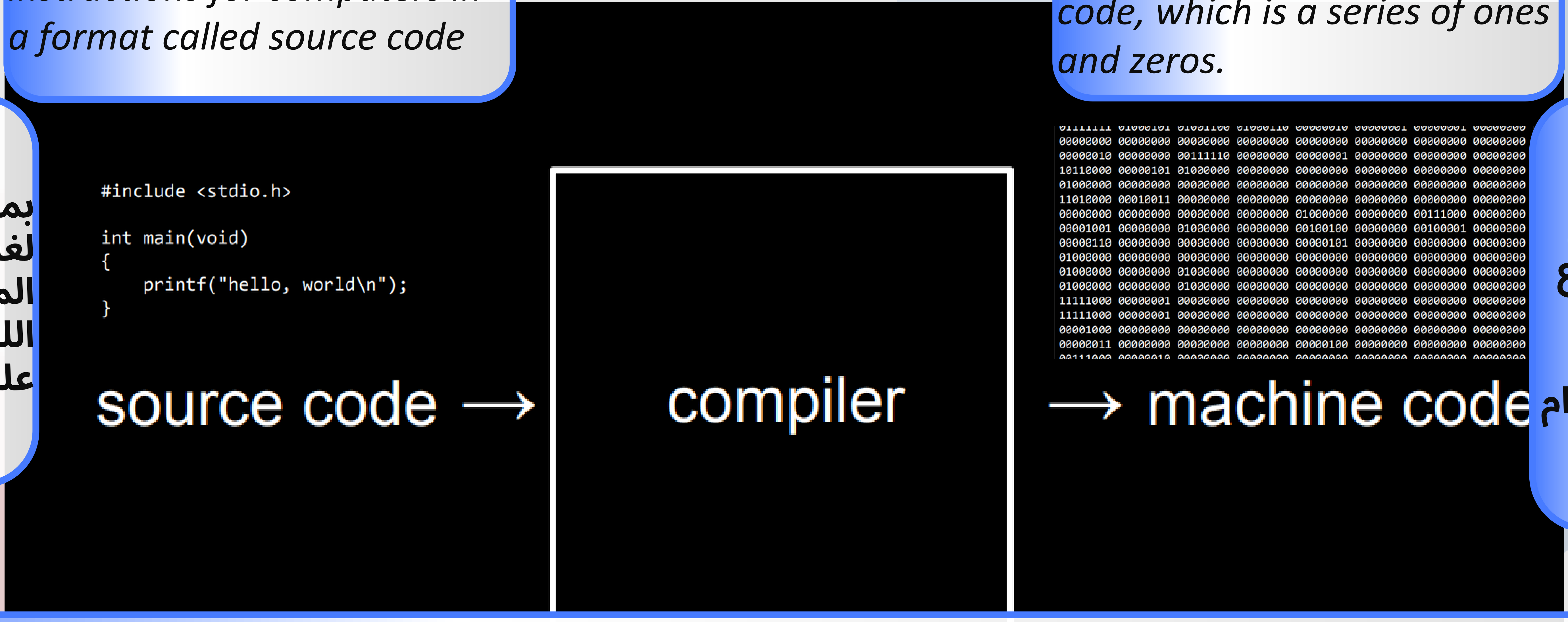
From Scratch to C++ programming:

Source Code: *Humans write instructions for computers in a format called source code*

VS

Machine Code: *Computers can only understand machine code, which is a series of ones and zeros.*

بما اننا هنبدأ نتعلم لغات برمجة فالكود المكتوب بكلمات اللغة syntax هنقول عليه source code



اتفقنا من week 0 ان الكمبيوتر بيفهم 0و1 ولكن الموضوع صعب علينا اننا نكتب للكمبيوتر التعليمات بإستخدام 0و1

فكنا محتاجين حل لان صعب علي البشر انهم يكتبوا machine code والكمبيوتر مبيفهمش غير 0و1 طيب الحل أي؟ الحل اننا نستخدم لغة قريبة من لغات البشر وسهله في قراءتها وفي نفس الوقت يكون في وسيط يفهم الكمبيوتر ويحوله اللغة الي احنا بنكتبها لmachine code زي ما المترجم بيعمل بالضبط طيب المترجم بالنسبة للغة البرمجة هنا اسمه compiler يبقي كدا نفهم ان وظيفة الcompiler انه يحول الأكواد السهلة الي بقدر اكتبها بحروف انجليزي للغة الي الكمبيوتر بيفهمها

Compiler: *A compiler is a tool that translates source code into machine code. In this session, we introduce a compiler for **C** programming.*

From Scratch to C++ programming:

High level language

الhigh level language هي اقرب للغات البشر زي ما شوفنا كدا syntax لغة ++C وبتكون محتاجة وسيط علشان يحول الsource code لmachine code يفهمه الكمبيوتر

Low level language

الlow level language زي الmachine code الكمبيوتر بيفهمها وبيقدر يتعامل معاه مباشر ولكن صعبة بالنسبة للبشر انهم يكتبوها ويتعاملوا بيها

First C++ program:

- *First of All, we will use Visual Studio Code.*
- *You can open **VS Code** at cs50.dev*
- *Or you can use Programiz directly.*
- *Next slide will be on how to use vs code to write your First C++ program.*


EXPLORER
HELLO [CODESPACES]
hello.c
hello.c
#include <stdio.h>
int main(void)
{
printf("hello, world\n");
}
This region in the middle called a **text editor** where you can edit your program
هنا text editor بنستخدمه علشان نكتب الأكواد
This is the **file explorer** bar on the left side where you can find your files
TERMINAL
$ make hello
OUTLINE
TIMELINE

First C++ program:

*We can build your first program in C++ by typing **code hello.cpp** into the terminal window. Notice that we deliberately lowercased the entire filename and included the **.cpp** extension. Then, in the text editor that appears, write code as follows:*

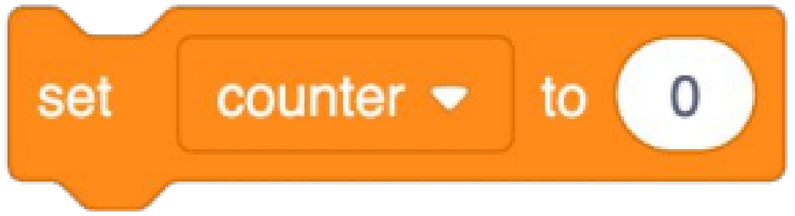
C++

```
#include "iostream"

using namespace std;

int main() {
    cout << "Hello, World!" << endl;
    return 0;
}
```

Scratch

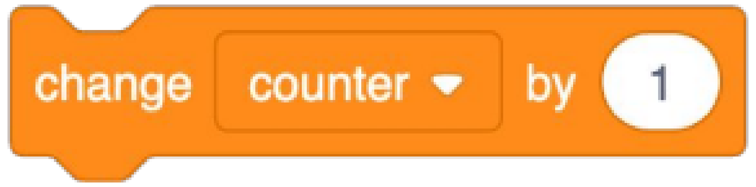
In **Scratch**, we utilized the **say** block to display any text on the screen. Indeed, in **C++**, we can use cout <<.

Variables:



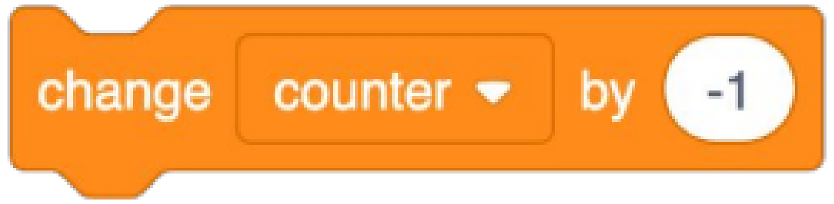
Initialization

```
int counter = 0;
```




```
counter = counter + 1;
```

```
counter += 1;

counter++;
```




```
counter--;
```

Declaration

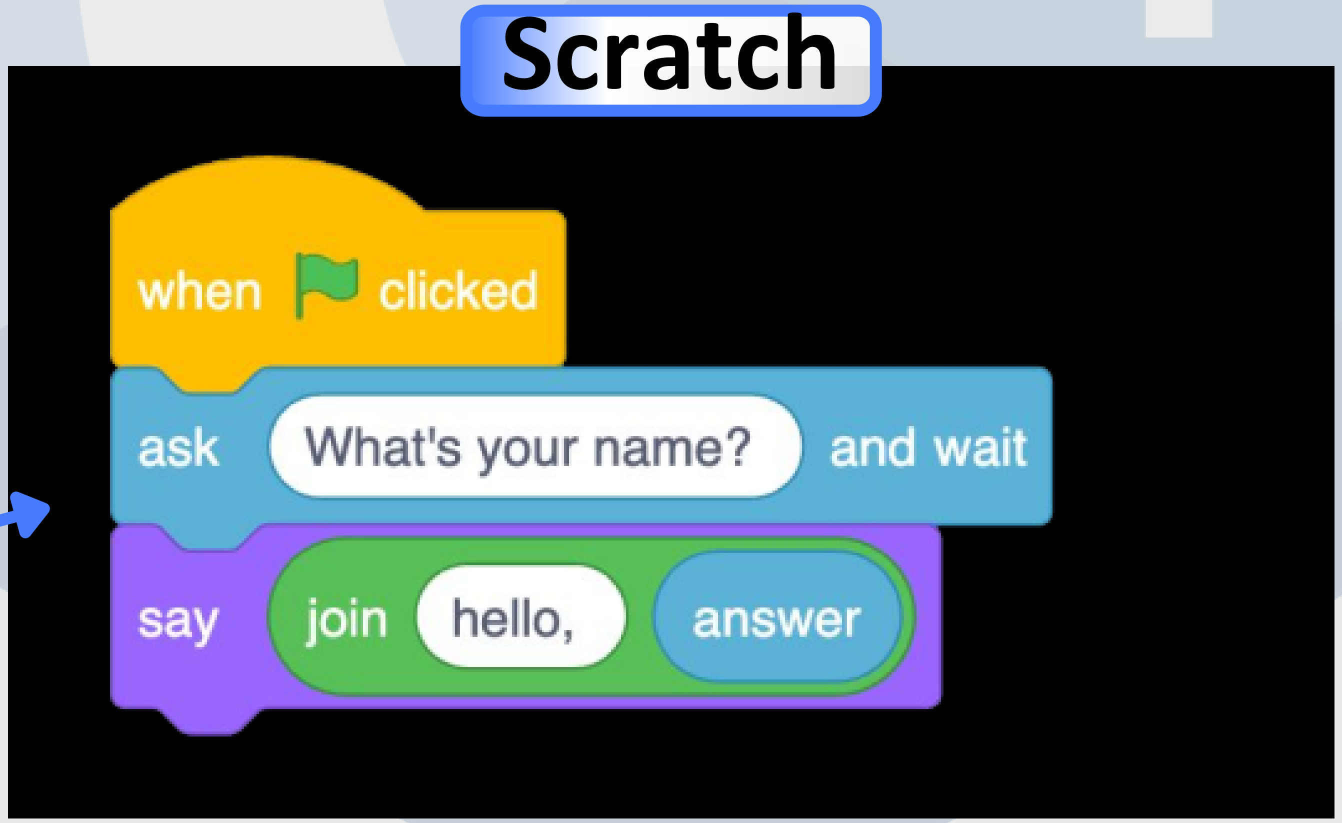
```
int counter;
```

```
bool
char
double
float
int
long
string
```

Variables:

Recall that in **Scratch**, we had the ability to ask the user **“What’s your name?”** and say **“hello”** with that name appended to it.
In **C**, we can do the same. Modify your code as follows:

```
#include <iostream>
using namespace std;

int main()
{
    string name;
    cout << "What's your name? ";
    cin >> name;
    cout << "Hello, " << name << endl;
    return 0;
}
```



Variables:

Given two numbers **X** and **Y**. Print the **summation** and **multiplication** and **subtraction** of these 2 numbers.

Input:
5 **10**
Output:
5 + 10 = 15 **5 * 10 = 50** **5 - 10 = -5**

Variables:

Implement a program that take the **Height** and **width** from the user and calculate the area of a rectangle.
Area of a rectangle = **height** * **width**

Input:
4 5
Output:
Area = 20

Variables:

Implement a program that take the **radius** from the user and calculate the area of a Circle and perimeter Law of area **=** suppose **= 3.14**

Law of perimeter **= 2**

Input:
Radius = 2.0
Output:
Area = 12.56 **Perimeter = 12.56**

Variables:

Implement a program that take the **radius** from the user and calculate the area of a Circle and perimeter. suppose PI = 3.14

Law of area **= PI * R * R**

Law of perimeter **= 2 * PI * R**

Input:
Radius = 2.0
Output:
Area = 12.56 **Perimeter = 12.56**

Variables:

Implement a program that take the **radius** from the user and calculate the area of a Circle and perimeter Law of area **=** suppose **= 3.14**

Law of perimeter **= 2**

Input:
Radius = 2.0
Output:
Area = 12.56 **Perimeter = 12.56**

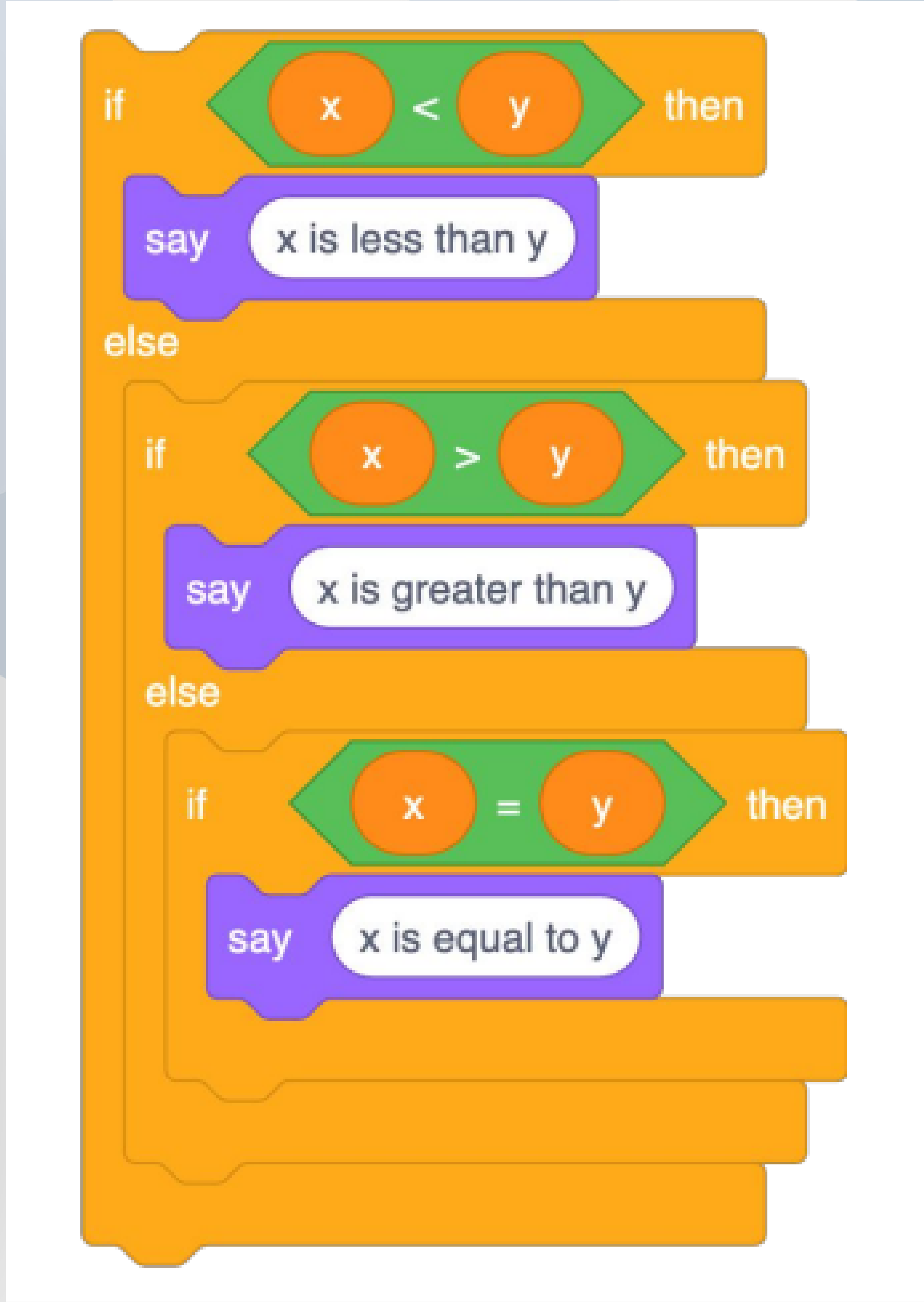
Boolean & Logical Operators:

```cpp
#include <iostream>
using namespace std;

int main()
{
    int x, y;
    cin >> x >> y;
    cout << (x < y) << endl;
    cout << (x > y) << endl;
    cout << (x <= y) << endl;
    cout << (x >= y) << endl;
    cout << (x == y) << endl;
    cout << (x != y) << endl;
    return 0;
}
```

```cpp
#include <iostream>
using namespace std;

int main()
{
    int x, y, z;
    cin >> x >> y >> z;
    cout << (x < y && z < y) << endl;
    cout << (x > y || z < y) << endl;
    cout << !(x <= y) << endl;
    return 0;
}
```

Conditionals:

```cpp
#include <iostream>
using namespace std;

int main()
{
    int x, y;
    cin >> x >> y;
    if (x < y) {
        cout << "x is less than y" << endl;
    } else if (x > y) {
        cout << "x is greater than y" << endl;
    } else {
        cout << "x is equal to y" << endl;
    }
    return 0;
}
```



Conditionals:

Implement a program that take to variable **X** and **Y** from the user and

if X less than Y print **"X is less than Y"**

if X greater than Y print **"X is greater than Y"**

if X equal Y print **"X is equal to Y"**

:Input
X = 5 Y = 10
:Output
X is less than Y

:Input
X = 10 Y = 5
:Output
X is greater than Y

:Input
X = 5 Y = 5
:Output
X is equal to Y

Conditionals:

Implement a program that take to variable **X** and **Y** from the user and

if X less than Y print **"X is less than Y"**

if X greater than Y print **"X is greater than Y"**

if X equal Y print **"X is equal to Y"**

```
#include <iostream>
using namespace std;

int main()
{
    int x, y;
    cin >> x >> y;
    if (x < y) {
        cout << "x is less than y" << endl;
    } else if (x > y) {
        cout << "x is greater than y" << endl;
    } else {
        cout << "x is equal to y" << endl;
    }
    return 0;
}
```

Conditionals:

Implement a program that takes a character from the user and
if this character is **'Y'** or **'y'** print **"YES“**
if this character is **'N'** or **'n'** print **"NO"**

Input
Y
Output
YES

Input
n
Output
NO

Input
v
Output
Invalid option!

Conditionals:

Implement a program that takes a character from the user and
if this character is **'Y'** or **'y'** print **"YES“**
if this character is **'N'** or **'n'** print **"NO"**

```cpp
#include <iostream>
using namespace std;

int main()
{
    char option;
    cin >> option;
    if(option == 'Y' || option == 'y') {
        cout << "YES" << endl;
    } else if(option == 'N' || option == 'n') {
        cout << "NO" << endl;
    } else {
        cout << "Invalid option!" << endl;
    }
    return 0;
}
```

Conditionals:

Implement a program that takes Guess character, and a number determine if his guess is correct or not Guess can be one of these characters **'P'** for positive numbers, **'N'** for negative numbers and **'Z'** for zero.
Output **“YES”** if his guess is correct and **“NO”** otherwise.

Input:
P **5**
Output:
YES

Input:
N **-1**
Output:
YES

Input:
P **-5**
Output:
NO

Input:
Z **0**
Output:
YES

Conditionals:

Implement a program that takes Guess character, and a number determine if his guess is correct or not Guess can be one of these characters **'P'** for positive numbers, **'N'** for negative numbers and **'Z'** for zero.
Output **“YES”** if his guess is correct and **“NO”** otherwise.

```cpp
#include <iostream>
using namespace std;

int main()
{
    int num;
    char guess;
    cin >> guess >> num;
    if (guess == 'P' && num > 0){
        cout << "YES" << endl;
    }else if(guess == 'N' && num < 0){
        cout << "YES" << endl;
    }else if(guess == 'Z' && num == 0){
        cout << "YES" << endl;
    }else{
        cout << "NO" << endl;
    }
    return 0;
}
```

Conditionals:

1. Given 3 numbers **A, B, C** print the **minimum** and **maximum** value.

Input:
A = 1 **B = 2** **C = 3**
Output:
Minimum = 1 and Maximum = 3

Input:
A = 1 **B = 1** **C = 1**
Output:
Minimum = 1 and Maximum = 1

Input:
A = 2 **B = 1** **C = 2**
Output:
Minimum = 1 and Maximum = 2

Conditionals:

1. Given 3 numbers **A, B, C** print the **minimum** and **maximum** value.

```cpp
#include <iostream>
using namespace std;

int main()
{
    int x, y, z;
    cin >> x >> y >> z;
    int minValue, maxValue;
    if (x < y && x < z) {
        minValue = x;
    } else if (y < x && y < z) {
        minValue = y;
    } else {
        minValue = z;
    }
    if (x > y && x > z) {
        maxValue = x;
    } else if (y > x && y > z) {
        maxValue = y;
    } else {
        maxValue = z;
    }
    cout << "Min: " << minValue << ", Max: " << maxValue << endl;
    return 0;
}
```

Support Team